FW Community Hub

Fort Wayne Community Resource Finder

Project Name	FW Community Hub
Version	1.0 - Internal Draft
Date	March 2026
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Duration	8 Weeks
Status	Pre-Development Planning

1. Project Overview

FW Community Hub is a free, publicly accessible web application that helps Fort Wayne, Indiana residents quickly find local community support resources. Instead of navigating confusing government websites or calling multiple phone numbers, users can search naturally or chat with an AI assistant that understands plain language like "I need food help tonight" and responds with real, verified local resources.

The Problem We Are Solving

Fort Wayne has 200+ community resources - food, housing, mental health, addiction recovery, job training, and more.

These resources are scattered across multiple websites, outdated PDFs, and phone directories.

Indiana 211 has a comprehensive database but it is difficult to use, not mobile-friendly, and has no AI or chat capability.

A person in crisis should not have to navigate complex systems to get help.

Our Solution

A clean, fast, mobile-friendly search and chat interface for Fort Wayne community resources.

An AI chat bot powered by Google Gemini that understands natural language.

An emergency mode that shows only 24/7 crisis resources instantly.

An admin panel for managing and verifying resource listings.

A public form for organizations to submit their own listings.

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2. Goals and Success Criteria

2.1 Primary Goals

- Build and deploy a working MVP within
- Seed the database with at least 30 verified Fort Wayne resources across all major categories
- Integrate an AI chat assistant that responds helpfully to natural language requests
- Make the app publicly accessible via a live URL on GCP
- Use the project as a portfolio piece to impress Fort Wayne tech employers

2.2 Secondary Goals

- Reach out to Indiana 211 and Fort Wayne nonprofits for a data partnership
- Apply for community funding to sustain hosting and API costs long term
- Grow the resource database to 200+ listings post-launch
- Add a map view and route planning in a future release

3. Users and Workflows

3.1 Public User - Fort Wayne Resident

This is the most important user. They may be in a difficult situation - hungry, struggling with housing, or in a mental health crisis. They need help fast and should never be required to create an account.

- Lands on home page
- Types a natural language query OR clicks a category button
- Views a list of matching resources with name, address, phone, and hours
- Clicks a resource to see full details and a map pin
- Clicks Get Directions to open Google Maps
- OR clicks I Need Help NOW to see only 24/7 crisis resources immediately
- OR opens the AI chat and describes their situation in plain language

3.2 Resource Organization

A nonprofit, church, or community group that wants to be listed in the app.

- Finds the app and clicks Add Your Organization
- Fills out a simple form with name, category, address, phone, hours, and languages
- Submits and receives a confirmation message
- Gets an email notification when their listing is approved and goes live

3.3 Admin - Developer or Trusted Volunteer

Manages data quality. Keeps listings accurate and up to date.

- Logs in at /admin using username and password
- Views a dashboard with total resources, pending approvals, and recent activity
- Reviews submitted resources and approves or rejects them
- Edits existing resources when information changes
- Deactivates resources for organizations that have closed

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4. Tech Stack and Architecture

4.1 Stack Overview

Layer	Technology	Reason
Backend	ASP.NET Core (C#)	Industry standard for Fort Wayne employers
Database	SQL Server	Fully managed on Azure, integrates natively with .NET
ORM	Entity Framework Core	Maps C# models to SQL Server tables
Frontend	Angular + Tailwind CSS	Fast to build
Auth	ASP.NET Identity + JWT	Industry standard for .NET APIs
AI Chat	Google Gemini API (free tier)	Free to start, easy to scale
Hosting	Azure App Service	Managed platform, no server maintenance
Database Hosting	Azure SQL Database	Managed SQL Server in the cloud
CI/CD	GitHub Actions	Automated build and deploy on every push to main

4.2 Architecture Diagram

The frontend is deployed as a static Angular build served via Azure Static Web Apps or from the App Service. The backend runs on Azure App Service. The database runs on Azure SQL Database (managed SQL Server).

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Azure Cloud
  ├── Azure App Service
  |     ├── ASP.NET Core API (handles /api/* routes)
  |     └── Serves Angular build (static files)
  └── Azure SQL Database (SQL Server)
External: Gemini API (Google AI)
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4.3 Key Design Decisions

Decision	Rationale
Azure App Service over a VM	No server maintenance. Microsoft manages OS, patching, and scaling. Easier for a student team.
SQL Server via Azure SQL Database	Native .NET integration. Managed backups, scaling, and security built in. No manual DB setup.
Gemini free tier for AI	No credit card needed. 250 req/day is enough for early launch. Easy to upgrade when users grow.
No login for public users	A person in crisis must never be blocked by a signup form. Zero friction to get help.
Repository pattern in C#	Separates database logic from controller logic. Matches what Fort Wayne employers expect.

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5. Features

5.1 MVP Features

Feature	Description	Priority
Search resources	Keyword search with real-time results	P0
Category filter	Filter by Food, Housing, Mental Health, etc.	P0
Resource detail page	Full info: address, phone, hours, languages	P0
Emergency mode	I Need Help NOW - shows only 24/7 resources	P0
AI chat bot	Natural language queries via Gemini API	P0
Admin login	JWT-secured admin access	P0
Admin dashboard	View, edit, deactivate resources	P0
Live deployment	Accessible via public GCP URL	P0

5.2 Post-Launch Features

Feature	Description	Priority
Submit a resource form	Public form for organizations to apply	P1
Pending approvals workflow	Admin reviews and approves submissions	P1
Map view	All resources shown as pins on a map	P1
Email notifications	Notify org when listing approved/rejected	P1
211 data integration	Import verified data from Indiana 211	P2
Multi-language support	Burmese, Spanish, English UI	P2
Analytics dashboard	Track most searched categories	P3

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6. Data Model

6.1 Core Resource Entity

Field	Type	Notes
Id	int	Auto-incremented primary key
Name	string	Organization name
Description	string	Short paragraph about the service
Category	string	Food, Housing, Mental Health, etc.
Address	string	Full street address
Latitude / Longitude	double	For map display
PhoneNumber	string	Primary contact number
Website	string	URL (optional)
HoursOfOperation	string	e.g. Mon-Fri 9am-5pm
Is24Hours	bool	Used for emergency mode filter
LanguagesSpoken	string	Comma-separated list
IsVerified	bool	Admin has confirmed accuracy
IsActive	bool	False = soft deleted
CreatedAt	DateTime	Auto-set on creation
UpdatedAt	DateTime	Auto-set on update

6.2 Resource Categories

- Food and Meals
- Housing and Shelter
- Mental Health
- Addiction and Recovery
- Job Training and Employment
- Financial Assistance
- Healthcare (free or low-cost)
- Family and Children Services
- Senior Services
- Veterans Services
- Transportation
- Legal Aid
- Domestic Violence
- Disability Services

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7. API Endpoints

7.1 Public Endpoints (No Login Required)

Method	Endpoint	Description
GET	/api/resources	Get all active resources (supports ?category= and ?search=)
GET	/api/resources/{id}	Get a single resource by ID
GET	/api/resources/emergency	Get only 24/7 crisis resources
GET	/api/categories	Get list of all categories
POST	/api/chat	Send a message to the AI chat bot
POST	/api/resources/suggest	Submit a new resource for review

7.2 Admin Endpoints (JWT Required)

Method	Endpoint	Description
POST	/api/auth/login	Admin login - returns JWT token
GET	/api/admin/resources	Get all resources including inactive
GET	/api/admin/pending	Get submitted but unverified resources
POST	/api/admin/resources	Create a new resource
PUT	/api/admin/resources/{id}	Update a resource
PUT	/api/admin/resources/{id}/approve	Approve a pending submission
DELETE	/api/admin/resources/{id}	Soft delete a resource

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8. 10-Week Development Plan

Weeks 1-2:

Week	Tasks
Week 1	
Week 2	

Weeks 3-4:

Week	Tasks
Week 3	
Week 4	

Weeks 5-6:

Week	Tasks
Week 5	
Week 6	

Weeks 7-8: CI/CD, Deployment, Polish, and Launch

Week	Tasks
Week 7	
Week 8	

9. Cost Breakdown

Item	Monthly Cost	Notes
Azure App Service (Free tier)	$0.00	F1 free tier - good for development and early launch
Azure App Service (Basic tier)	~$13.00	B1 plan - needed for custom domain and always-on
Azure SQL Database	~$5.00	Basic tier, 2GB storage, plenty for early use
Gemini API (free tier)	$0.00	250 requests/day - sufficient for early launch
Domain name (optional)	~$1.00	$12/year for fwresourcehub.org
TOTAL (free tier)	~$1/month	App Service F1 + free Gemini + optional domain
TOTAL (basic tier)	~$19/month	App Service B1 + Azure SQL + optional domain

Azure for Students

Purdue Fort Wayne students qualify for Azure for Students, which gives $100 in free Azure credits per year.

This covers approximately 5-7 months of Basic tier hosting with no credit card required.

Sign up at azure.microsoft.com/free/students using your Purdue email address.

If Gemini free tier is exceeded, Gemini paid tier costs approximately $0.003 per conversation - very affordable at low usage.

10. Future Opportunities

10.1 Partnership Targets

- Indiana 211 - data sharing partnership for their full verified resource database
- City of Fort Wayne Department of Neighborhoods - official recognition or micro-grant
- Northeast Indiana Innovation Center (NIIC) - mentorship and funding connections
- Allen County Department of Health - healthcare resource accuracy partnership
- HART (Hope and Recovery Team) - addiction recovery resource verification

10.2 Funding Sources

- City of Fort Wayne CDBG grants - via a nonprofit partner organization
- Community Foundation of Greater Fort Wayne - civic engagement grants ($5,000-$300,000)
- Sparked at Inspire micro-grant - new program from Indiana Michigan Power Foundation
- Google Fort Wayne Community Grants - Google has given $500,000+ to local nonprofits

10.3 Potential Enhancements

- Burmese language support - serve Fort Wayne's Burmese community directly
- SMS integration - text a ZIP code and get resources back via text message
- Volunteer coordination - let users sign up to volunteer at listed organizations
- Resource ratings - let verified users flag outdated or incorrect listings

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Definition of Done for MVP

All P0 features are working on the live Azure App Service URL.

CI/CD pipeline is active: every push to main triggers an automated build and deploy via GitHub Actions.

At least 30 verified Fort Wayne resources are seeded and searchable.

AI chat correctly handles at least: food, housing, mental health, and addiction queries.

Admin can log in, view all resources, and edit or deactivate them.

App works correctly on a mobile phone browser.

GitHub README is complete with screenshots and live URL.

Live URL is added to both developers' portfolio websites.